

AIGR Deep Research — *S. pombe* iwr1 (O13951) core-function hypothesis

Hypothesis under review: *Schizosaccharomyces pombe* Iwr1 is a conserved RNA polymerase II nuclear-import adaptor that binds assembled Pol II and presents its own NLS to karyopherin- α .

Focus type: core_function · **Gene:** iwr1 / SPAC23H4.08 · **UniProt:** O13951 (277 aa)

Executive Judgment

Verdict: Partially supported (supported by orthology; not experimentally demonstrated in *S. pombe*).

The described activity — an Iwr1 protein that binds assembled Pol II and presents its own N-terminal bipartite NLS to karyopherin- α to drive Pol II nuclear import — is a *bona fide*, directly demonstrated **core molecular function of the *Saccharomyces cerevisiae* ortholog** (Czeko et al., 2011,

P 21504834

), and the authors explicitly report it is "apparently conserved from yeast to human." For the *S. pombe* target specifically, the assignment is **entirely inference-based**: every GO annotation on O13951 is electronic/phylogenetic (IEA/ISO/IBA), the molecular-function slot is **ND (no data)**, and no primary *S. pombe* experimental study of iwr1 exists in the literature. Sequence analysis I ran supports transferability — O13951 carries the Iwr1 Pfam domain (PF08574) and conserves the N-terminal basic (NLS-like) clusters and acidic C-terminal tail seen in budding-yeast Iwr1 — but the NLS / karyopherin- α interaction and Pol II binding have **not been tested in fission yeast**.

Most important caveats. (1) The "presents its own NLS to karyopherin- α " clause is a *S. cerevisiae* mechanistic detail, transferred by conservation, not verified in *S. pombe*. (2) Iwr1 is **non-essential** and Pol II import can proceed by Iwr1-independent routes (P 26455955),

so it is a **facilitating adaptor, not the obligate importer**. (3) Budding-yeast Iwr1 has additional documented roles (elongation, Pol II disassembly from chromatin, genome stability, gene-specific transcription); the import-adaptor role is the best-supported **direct** function, whereas transcription phenotypes are largely downstream. (4) Ortholog identity is **low** (~23% to *S. cerevisiae*, ~20% to human SLC7A6OS); orthology rests on the shared Iwr1 Pfam domain (PF08574) and positionally conserved N-terminal basic clusters rather than high sequence identity.

Supporting feasibility checks I ran this iteration. The AlphaFold model of O13951 is largely low-confidence/disordered (mean pLDDT 64.6) but the **N-terminal NLS cluster (res 5–15) is well-ordered (pLDDT ≈91)** — structurally consistent with an intrinsically disordered Pol II-cleft adaptor carrying a defined, accessible NLS. *S. pombe* also **encodes the required importin- α receptors** (imp1/O94374, cut15/O14063), so the proposed karyopherin- α -dependent mechanism is feasible in this organism (though the Iwr1–importin- α interaction itself is untested here).

Evidence Matrix

Citation (PMID)	Evidence type	Stance	Claim tested	Key finding	Context	
21504834 (Czeko 2011)	Direct assay / structural / interaction	Supports	Iwr1 binds assembled Pol II and presents own NLS to karyopherin- α for import	Iwr1 binds Pol II in the active-center cleft between the two largest subunits; N-terminal bipartite NLS recognized by karyopherin-α directs import; displaced by initiation factors/ nucleic acids; "Pol II specific, transcription independent, apparently conserved from yeast to human."	<i>S. cerevisiae</i> (+ conservation claim)	F C c t a r s
19679657 (Peiró-Chova 2009)	Mutant phenotype / interaction	Qualifies	Iwr1 role in transcription vs import	Iwr1p physically interacts with most Pol II subunits; genetic interactions with basal machinery; nucleocytoplasmic shuttling; affects specific-gene transcription	<i>S. cerevisiae</i>	T e d c a
26455955 (Gómez-Navarro 2015)	Mutant phenotype / localization	Qualifies	Iwr1 is the sole Pol II importer	IWR1 is non-essential; Iwr1-independent import pathways exist (holoenzyme, partial assemblies, diffusion of small subunits)	<i>S. cerevisiae</i>	I f a c s " v i
28258010 (Gómez-Navarro 2017)	Mutant phenotype / genetics	Competing/ qualifies	Import is Iwr1's only role	Iwr1 also acts during elongation and in Pol II disassembly from chromatin ; iwr1 Δ	<i>S. cerevisiae</i>	A a c

Citation (PMID)	Evidence type	Stance	Claim tested	Key finding	Context	
				causes genome instability, genotoxin sensitivity		
23267056 (Minaker 2013)	Mutant phenotype / localization	Supports (pathway)	Iwr1 in Pol II (and III) biogenesis	iwr1Δ Pol II import defect is suppressed by an NLS fused to Rpb3, consistent with Iwr1 supplying the NLS; GPN GTPases act upstream	<i>S. cerevisiae</i>	
23438601 (Gómez-Navarro 2013)	Interaction / pathway	Qualifies	Import machinery composition	Rtp1/karyopherin-like factors and R2TP participate in Pol II assembly/import alongside Iwr1	<i>S. cerevisiae</i>	
UniProt O13951 (by similarity)	Database	Supports (orientation)	Function of <i>S. pombe</i> protein	Annotated "Directs RNA polymerase II nuclear import," "Associates with RNA polymerase II," Cytoplasm+Nucleus, IWR1/SLC7A6OS family (PF08574)	<i>S. pombe</i>	
QuickGO annotations for O13951	Database	Qualifies	Current curation state	BP GO:0006606 (IEA/ISO/IBA); CC GO:0005634 + GO:0005737 (ISO/IBA/IEA); MF = GO:0003674 ND	<i>S. pombe</i>	
This report (computed)	Computational / evolutionary	Supports	Architecture conserved in O13951	Iwr1 domain present; N-terminal basic clusters (RVKRRK res7–11, TKRRK res34–38) match budding-yeast	<i>S. pombe</i> vs <i>S. cerevisiae</i> Q07532	

Citation (PMID)	Evidence type	Stance	Claim tested	Key finding	Context	
				bipartite-NLS position; acidic C-terminal tail conserved		
This report (computed, AlphaFold DB)	Structural / computational	Supports	Is the NLS in an ordered, accessible element?	O13951 is mostly low-confidence/disordered (mean pLDDT 64.6), but N-terminal NLS cluster1 (res 5–15) is well-ordered (pLDDT ~91); rest of protein and acidic C-tail are low-pLDDT	<i>S. pombe</i> AlphaFold model	M c a c h c t
This report (computed, NW alignment)	Evolutionary / computational	Qualifies	Strength of orthology	Global identity O13951 vs <i>S. cerevisiae</i> Iwr1 = 22.7% ; vs human SLC7A6OS = 20.4% — low; orthology rests on shared Pfam PF08574 + positionally conserved basic clusters, not high identity	pairwise	I v r t f a t s
This report (computed, UniProt)	Database / feasibility	Supports (feasibility)	Does the required receptor exist in <i>S. pombe</i> ?	<i>S. pombe</i> encodes importin- α orthologs imp1 (O94374) and cut15 (O14063)	<i>S. pombe</i>	P r n r I i h u P
16823372 (Matsuyama 2006)	Localization (high-throughput)	Supports (CC)	Subcellular localization in fission yeast	Genome-wide YFP localization of ~90% of <i>S. pombe</i> proteome (resource that	<i>S. pombe</i>	P t i c

Citation (PMID)	Evidence type	Stance	Claim tested	Key finding	Context	
				provides direct <i>S. pombe</i> localization data)		v P c
PomBase SPAC23H4.08 (record)	Database	Qualifies	Is there direct <i>S. pombe</i> evidence?	characterisation_status = " biological role inferred "; deletion viable; single copy (1:1) , no paralog; GO = CC cytoplasm+nucleus, BP protein import into nucleus, no MF term ; descriptor "nucleocytoplasmic shuttling protein"; GO transfer via GO_REF:0000033 (IBA)/0000024 (ISO)	<i>S. pombe</i>	C i e s c P c
PomBase phenotypes (screens: P 20473289 P 37787768)	Mutant phenotype (high-throughput)	Qualifies / competing	Are loss phenotypes the core function?	~30 pleiotropic screen phenotypes: drug/stress resistance (MMS, HU, caffeine, cadmium, cycloheximide), decreased mating efficiency, multiseptate, loss of stationary-phase viability	<i>S. pombe</i> deletion screens	P c n i P c S g s (— c s

GO Curation Implications (leads — require curator verification)

- **BP GO:0006606 "protein import into nucleus" — RETAIN.** Well-supported by orthology (ISO from SGD IWR1, IBA, IEA) and consistent with conserved architecture. It is appropriately general; there is no granular "RNA polymerase II complex nuclear import" GO term, so 0006606 is the correct level. Evidence code should remain non-experimental (ISO/IBA) for *S. pombe*.
 - **MF is currently ND — a candidate ADD (lead).** The hypothesis implies a definable molecular function as an **import adaptor**. Best-supported candidate MF terms, by orthology only:
 - **GO:0030674 "protein-macromolecule adaptor activity"** — captures bridging Pol II cargo to karyopherin- α ; the most informative MF consistent with the mechanism.
 - **GO:0000993 "RNA polymerase II complex binding"** — captures the direct Pol II association.
 - Avoid "protein binding" (GO:0005515) as a terminal recommendation; the above are more informative.
 - Because *S. pombe* evidence is inference-only, any MF addition should carry ISO/IBA evidence, not experimental codes.
 - **CC GO:0005634 (nucleus) + GO:0005737 (cytoplasm) — RETAIN.** Consistent with a shuttling adaptor and with the *S. pombe* ORFeome localization resource
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verify the iwr1-specific call).
 - **Do not over-state.** Avoid annotating iwr1 as the obligate Pol II importer or asserting a verified karyopherin- α interaction in *S. pombe*; the import pathway is redundant
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GO Decision Table (leads — require curator verification)

Aspect	Term	Current state (O13951)	Recommended action	Suggested evidence code	Rationale
BP	GO:0006606 protein import into nucleus	Present (IEA/ISO/IBA)	Retain	ISO/IBA	Correct granularity (no Pol-II-specific import term exists); well-supported by orthology; PomBase concurs
MF	GO:0030674 protein-macromolecule adaptor activity	Absent (MF = ND)	Consider ADD (lead)	ISO/IBA	Captures the cargo↔karyopherin- α bridging that is the essence of the hypothesis
MF	GO:0000993 RNA polymerase II complex binding	Absent	Consider ADD (lead)	ISO/IBA	Captures direct Pol II association ("binds assembled Pol II")
MF	GO:0005515 protein binding	—	Avoid as terminal	—	Uninformative; use the two MF terms above instead
CC	GO:0005634 nucleus	Present	Retain	ISO/IBA/IEA	Shuttling adaptor; consistent with <i>S. pombe</i> record
CC	GO:0005737 cytoplasm	Present	Retain	ISO/IBA/IEA	Cytoplasmic Pol II binding / recycling
BP/CC	genotoxin sensitivity, mating, drug-resistance phenotypes	PomBase screen phenotypes	Do NOT annotate as core function	—	Pleiotropic, downstream, screen-derived; some conflict with <i>S. cerevisiae</i>

All *S. pombe* MF/BP/CC calls should carry **non-experimental (ISO/IBA)** evidence — no experimental (EXP/IDA/IPI) code is justified, as no direct fission-yeast assay exists.

Mechanistic Scope

Direct molecular function under test: cytoplasmic binding of the assembled 12-subunit Pol II in its cleft, and presentation of Iwr1's own N-terminal bipartite NLS to karyopherin- α (importin- α ; *S. pombe* Cut15/Imp1 family) to mediate Pol II nuclear import, followed by Iwr1 displacement, export and recycling. This is a **transcription-independent** transport/adaptor activity.

Downstream / non-core effects (should not be conflated with the core function): gene-specific transcription changes, NC2/TFIIA genetic bypass, elongation and Pol II chromatin-disassembly roles, genome instability and genotoxin sensitivity in null mutants. These are documented for budding yeast and are largely **consequences of** (or roles adjacent to) the import/assembly function rather than the primary molecular activity.

Conflicts and Alternatives

- **Organism gap (principal issue):** all direct evidence is *S. cerevisiae*/human-family; the *S. pombe* claim is orthology transfer. This is the central caveat, not a contradiction.

- **Redundancy:** Iwr1-independent import pathways exist

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"adaptor" is accurate, "the importer" is too strong.

- **Multifunctionality:** Gómez-Navarro 2017

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supports additional direct roles (elongation, Pol II turnover), so "nuclear-import adaptor" may be **one of several** direct functions rather than the sole one — though it remains the best-characterized primary activity.

- **Pol specificity:** Minaker 2013

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reports iwr1 mutant Pol **III** localization defects too, mildly complicating the strict "Pol II-specific" clause.

- **Paralog/family note:** family is IWR1/SLC7A6OS (human SLC7A6OS); PomBase lists iwr1 as "**predominantly single copy (one to one)**" with no *S. pombe* paralog, so paralog over-annotation is unlikely here.
- **Phenotype direction conflict:** *S. pombe* iwr1Δ shows **resistance** to MMS/HU in genome-wide screens, whereas *S. cerevisiae* iwr1Δ is **sensitive** to genotoxins

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Likely an organism difference or high-throughput screen artifact; either way these DNA-damage phenotypes are peripheral to the core import-adaptor function and should not drive the core-function call.

- **Taxonomic-scope nuance:** PomBase tags iwr1 both "conserved in eukaryotes" and "conserved in fungi only"; a clear 1:1 ortholog is fungal, while the broader Pfam family reaches metazoa (human SLC7A6OS) only at ~20% identity. The seed's "conserved... yeast to human" is defensible at the family level but weak at the sequence-identity level.

Knowledge Gaps

1. **No direct *S. pombe* assay** of Iwr1–Pol II binding or Iwr1–karyopherin-α (Cut15/Imp1) interaction. *Checked:* PubMed (no iwr1 fission-yeast primary papers), UniProt (by-similarity), QuickGO (no EXP codes). *Matters:* the entire core-function call is inference. *Resolve:* co-IP/affinity capture of Pol II with tagged Iwr1 in *S. pombe*; NLS-deletion import assay.
2. **Functional NLS not verified in O13951.** *Checked:* sequence — basic clusters present in the conserved position. *Matters:* the "own NLS" clause is the crux of the hypothesis. *Resolve:* mutate res 7–11/34–38 and assay Rpb1 nuclear localization; test suppression by Rpb3-NLS fusion.
3. **iwr1-specific localization in *S. pombe* — partially resolved.** *Checked:* PomBase records CC cytoplasm + nucleus and the descriptor "nucleocytoplasmic shuttling protein," and flags characterisation_status = "biological role inferred." *Remaining:* these lack an experimental evidence code tied to a dedicated iwr1 study; confirm the ORFeome

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call and whether shuttling was directly observed for iwr1 in *S. pombe*.

4. **Essentiality / redundancy in *S. pombe*** unquantified. *Resolve*: iwr1Δ phenotype and Rpb1-GFP localization in fission yeast.

Discriminating Tests

1. **Iwr1–Pol II co-purification in *S. pombe*** (TAP/FLAG-Iwr1 → MS): confirms cargo binding.
2. **NLS-swap / NLS-deletion import assay**: delete O13951 N-terminal basic clusters, score Rpb1-GFP nuclear import; rescue by fusing a heterologous NLS to Rpb3 (mirrors

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3. **Iwr1–karyopherin-α (Cut15) two-hybrid / pulldown** in *S. pombe*.
4. **Structure prediction of the O13951–Pol II cleft interface** (AlphaFold-Multimer of Iwr1 with Rpb1/Rpb2) to test cleft-binding conservation *in silico*.
5. **Cross-complementation**: does *S. pombe* iwr1 rescue *S. cerevisiae* iwr1Δ import defect?

Curation Leads (require curator verification)

- **Action**: Keep BP GO:0006606; consider **adding an MF** (GO:0030674 protein-macromolecule adaptor activity and/or GO:0000993 RNA polymerase II complex binding) with **ISO/IBA** evidence, filling the current ND slot. Keep CC nucleus + cytoplasm.
- **Reference to cite for the mechanism (ortholog)**: PMID 21504834 — snippet to verify: "*Iwr1 then uses an N-terminal bipartite nuclear localization signal that is recognized by karyopherin α to direct Pol II nuclear import*" and "*Iwr1 function is Pol II specific, transcription independent, and apparently conserved from yeast to human.*"
- **Qualifier reference (redundancy)**: PMID 26455955 — establish that import is not Iwr1-obligate; annotate as adaptor/facilitator, not sole importer.
- **Localization support (fission yeast)**: PMID 16823372 (verify iwr1 call).
- **Suggested question for curator**: Should the *S. pombe* MF/BP additions be evidence-coded ISO/IBA (recommended) given the absence of any *S. pombe* experimental data?

- **Suggested experiments:** the discriminating tests above, especially the NLS-deletion import assay and Iwr1–Pol II co-IP in *S. pombe*.
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Provenance: UniProt REST (O13951, Q07532), QuickGO annotation API, and an executed sequence-conservation comparison (Iwr1 domain PF08574; N-terminal basic clusters; acidic C-tail) were run during this review. All *S. pombe*-level functional statements are inference from orthology; no *S. pombe* experimental primary literature was found.

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